

Precision vs Recall: Checklist & Thumb Rules

Scenario-Based Recommendations

- Spam Detection: Focus on Precision to avoid false positives.
- Disease Diagnosis: Focus on Recall to avoid missing actual cases.
- Credit Card Fraud: Balance both Precision & Recall.
- Search Engines: High Precision for quality results.
- Face Recognition: High Precision to avoid unlocking for wrong person.
- Criminal Identification: High Recall to catch all suspects.
- Ad Targeting: High Precision to target right users.
- Obstacle Detection: High Recall to avoid missing any danger.

General Thumb Rules

- Prefer Precision when false positives are costly.
- Prefer Recall when false negatives are dangerous.
- Use F1 Score when you want a balance.

Techniques to Improve Precision and Recall

- Use Data Balancing (SMOTE, oversampling, undersampling).
- Tune decision thresholds.
- Clean noisy or mislabeled data.
- Engineer better features using domain knowledge.
- Evaluate with Confusion Matrix and PR Curve.
- Use cross-validation and ensemble methods.

Tips to Explain to Students

- Use real-life analogies (spam vs. cancer detection).
- Use PR Curve and Confusion Matrix visualizations.
- Ask: Is it worse to raise a false alarm or miss a real case?